MMET 410 – Lab 2

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Lab report format

- Title Page – Lab Number, Name and Date
  - The body of the report should include the following for each task:
  - a. Problem Statements – Task Number
  - b. Sequence of operations
  - c. Input and output devices and terminal address assignments
  - d. Ladder logic program

Task 1

Task #1
A doorbell can be turned On and Off using two normally open **spring return push buttons**. One is located at the front door; the other one is located at the back door. If **either** button is pushed, the doorbell is activated. The push buttons are connected to **IN-0** and **IN-1** in the AC input module located at slot 8. The doorbell is activated by **OUT-0** in the Relay Output module located at slot 9.

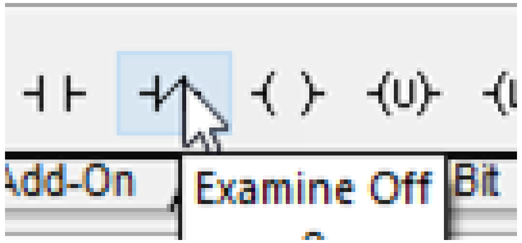
- 2 input :
  Button1: AC at slot 8 input IN-0
  Button2: AC at slot 8 input IN-1
- 1 output: Doorbell Relay Output at slot 9 output OUT-0
- When either of the push buttons is pressed, the doorbell will be true, so 2 inputs should be in parallel.

Task 2

Task #2
A motor can be turned on and off using a **normally closed switch (N.C.)**. When the switch is in the **Start** position (switch is **open**), the **motor will run.** When the switch is in the **Stop** position (switch is **closed**), the motor will **stop running**. The switch is connected to **IN-5** in the AC input module located at slot 8. The motor is activated by **OUT-0** in the Relay Output module located at slot 9.

- 1 input : Switch
- 1 output : Motor

Examine off means to examine if the bit is zero; if the circuit is open; the bit is zero, and the instruction is TRUE.



Example on and example off

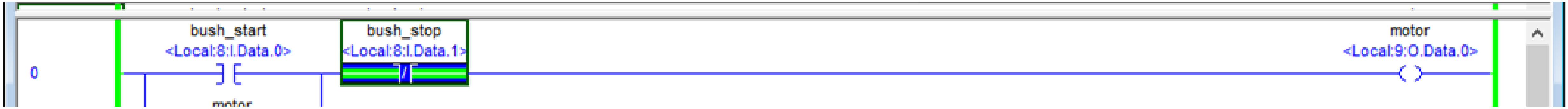
Example condition: if condition is satisfied, the component will become True / Green

Example on : Test if the target is on / 1 / close

Example off: Test if the target is off / 0 / open

Example (Task 3)

- The XIC and XIO instruction are assigned to start and stop button respectively.
- Step 1 - If push_start is pressed, the motor is energized and will remain on. If push_stop is pressed, the motor will turn off.
  - push_start Local:8:I:Data.0
  - push_stop Local:8:I:Data.1
  - motor Local:9:O: Data.0



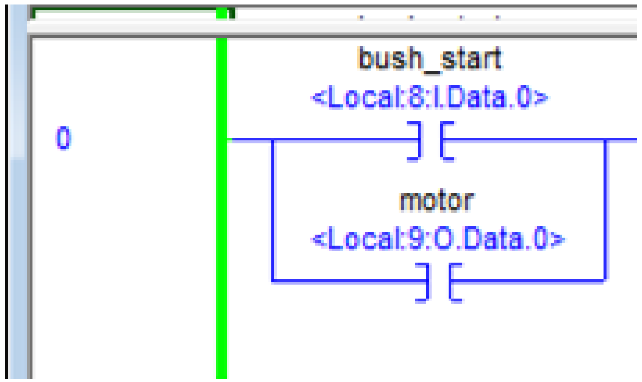
Task 3

Task #3

Most large industrial machines are **turned off** by using **separate spring push buttons** for ON and OFF (see figure below). This has safety implications in that the OFF push button(s) can be given priority to shut down the machines in an emergency regardless of the status of the ON push button. The **ON push button** is **normally open spring return** and **OFF** is **normally open spring return**. Please write a ladder diagram to use the ON and OFF buttons to control a motor. When the **ON** button is **pressed and released**; the **motor starts**. Once the motor starts, it will **continue until the OFF button is pressed**.

- 2 input : One input is examine-on, one input is examine-off
- 1 output

You assign the output address to examine on/off instruction to keep the output energized without having to press the input push button



ON button: IN-0 , OFF button: IN-1

Motor: OUT-0

Task 4

Task #4
A milling machine (M) and its lubrication pump (L) both have start/stop control systems. **L must be running before M can be started**. Furthermore, if **L stops, M must also stop**. Normally open spring return push buttons **StartM** and **StopM** are used to control the **machine(M)**. Normally open spring return push buttons **StartP** and S**topP** are used to control the **pump(L)**. Please assign necessary corresponding addresses for the inputs and outputs.

- 4 input
- 2 output

This is exactly the same as task 3

- Use IN-0 for **(L) ON** button, IN-1 for **(L) OFF** button
- Use IN-2 for **(M) ON** button, IN-3 for **(M) OFF** button
- Use OUT-0 for **(L)** , OUT-1 for **(M)**

Task 5

Task #5
A console is designed to control a motor which will pump water from the ground and then store the water in a tank for an industrial application. Water will be released from the tank if it is full. This console has only one control, an **ON/OFF switch**. The ON/OFF switch can be manipulated regardless of the state of the lock, but the function of the **ON/OFF switch is disabled if the lock is locked**. For instance, if the motor is running and the lock is changed to the locked position, turning the ON/OFF switch to OFF will have no effect. On the other hand, if the **lock is unlocked** to begin with; you can turn the ON/OFF switch to ON position to turn on the motor. Both ON/OFF and Lock/Unlock are normally open switches. Note: You will be using **two normally open switches** in the PLC box to mimic the ON/OFF and Lock/Unlock switches. Therefore, **Lock** means the **switch is in the ON** position and **Unlock** means the **switch is in the OFF** position.

- 2 input : Key switch (IN-4 slot 8), On/Off switch (IN-5 slot 8)
- 1 output: Motor (OUT-9 slot 9)

Task 5

Hint:
When the key switch is OFF (unlock):
on/off switch on, Motor on, on/off switch off, Motor off

When the key switch is ON (lock):
motor will **remain its current states**. Ignore the change of on/off switch.

- 2 input : Key switch (IN-4 slot 8), On/Off switch (IN-5 slot 8)
- 1 output: Motor (OUT-0 slot 9)

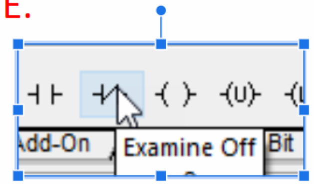
Task 1

- 2 input :
  Button1: AC at slot 8 input In-0
  Button2: AC at slot8 input IN-1
- 1 output: Doorbell Relay Output at slot 9 output OUT-0
- When either of the push buttons is pressed, the doorbell will be true, so 2 inputs should be in parallel.

Task 2

- 1 input : Switch
- 1 output : Motor

Examine off means to examine if the bit is zero; if the circuit is open; the bit is zero, and the instruction is TRUE.



Task 3

- 2 input
- 1 output

One input is examine-on

one input is examine-off

You assign the output address to examine on/off instruction to keep the output energized without having to press the input push button.

ON button: IN-0

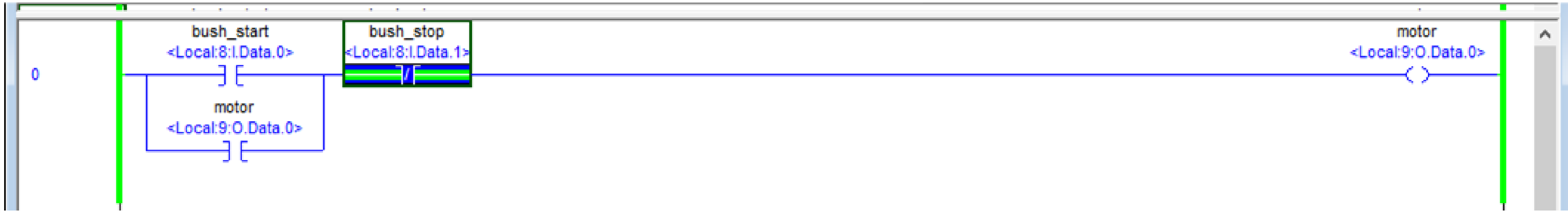
OFF button: IN-1

motor: OUT-0

Task 4

- 4 input
- 2 output

This is exactly the same as task3



Use IN-0 for **(L) ON** button

- Use IN-1 for **(L) OFF** button
- Use IN-2 for **(M) ON** button
- Use IN-3 for **(M) OFF** button
- Use OUT-0 for **(L)**
- Use OUT-1 for **(M)**

Task 5

Hint:
When the key switch is OFF (unlock):
on/off switch on, Motor on, on/off switch off, Motor off

When the key switch is ON (lock):
motor will **remain its current states**. Ignore the on/off switch change.

- 2 input
- 1 output

- Key switch : IN-4 slot 8
- on/off switch: IN-5 slot 8
- Motor: OUT-0 slot 9